Selected Solutions from Tutorials

Adapted from **Shigley, Problem 13-16** (8th Ed., 13-14).

The double-reduction gearset shown in the figure is driven through shaft *a* at a speed of 900 rev/min. Gears 2 and 3 are spur gears and have a diametral pitch of 10 teeth/in and a pressure angle of 20°. The second pair of gears in the train, gears 4 and 5, are helical gears with a normal diametral pitch of 6 teeth/in, a 25° helix angle, and a normal pressure angle of 20° (see Figure 13-22). The tooth numbers for the four gears are: $N_2 = 14$, $N_3 = 54$, $N_4 = 16$, and $N_5 = 36$.

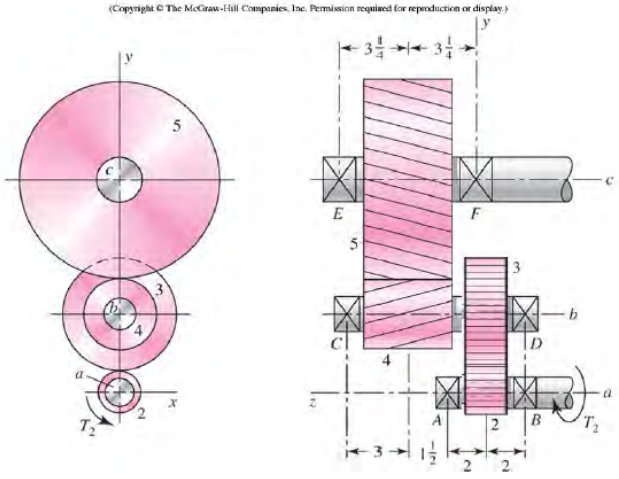
Part 1 (To be demonstrated by the TA). For the first stage (the spur gears) find:

d. The speed and direction of shaft *b*
e. The centre distance between shafts *a* and *b*

Part 2 (To be done by students). For the second stage (the helical gears) find:

f. The speed and direction of shaft *c*
g. The centre distance between shafts *b* and *c*
h. The directions of the thrust forces exerted by gears 4 and 5 on shafts *b* and *c*, respectively

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2. Adapted from Shigley, Problem 13-16.

a) Find the speed and direction of shaft *b* (same as speed and direction of gear 3)

$$n_3 = e_{23} n_2 = -\frac{\text{driving teeth}}{\text{driven teeth}} n_2$$

$$= -\frac{N_2}{N_3} n_2 = -\frac{14}{54}(-900)$$

$$= 233 \text{ rev/min (CW)}$$

b) Find the centre to centre distance between the shafts *a* and *b*.

When properly spaced, pitch circles of gears 2 and 3 are in contact. The shaft to shaft distance is half of the sum of the two pitch circle diameters.

$$P = \frac{N}{d} \qquad \text{(Equation 13-1)}$$

where

P = diametral pitch (number of teeth per inch)
N = number of teeth
d = diameter (inch)

$$d_2 = \frac{N_2}{P} = \frac{14 \text{ teeth}}{10 \text{ teeth/in}} = 1.4 \text{ inch}$$

$$d_3 = \frac{N_3}{P} = \frac{54 \text{ teeth}}{10 \text{ teeth/in}} = 5.4 \text{ inch}$$

Shaft-to-shaft distance, y_{23}, = $\frac{d_2 + d_3}{2} = \frac{1.4 + 5.4}{2} = 3.4$ inch

c) Find the speed and direction of shaft *c* (same as speed and direction of gear 5).

Note: gear 3 and gear 4 are on common shaft → $n_3 = n_4$

$$n_5 = e_{45} n_4 = -\frac{\text{driving teeth}}{\text{driven teeth}} n_4$$
$$= -\frac{N_4}{N_5} n_4 = -\frac{16}{36}(-233)$$
$$= 103.6 \text{ rev/min(CCW)}$$

d) Find the centre to centre distance between the shafts *b* and *c*.

We are given normal diametral pitch (measured perpendicular to teeth). For a helical gear, this needs to be converted to transverse diametral pitch (measured around the circumference). See Fig 13-22 and Eq. 13-18 in text.

$$P_{transverse} = P_{normal} \cos\psi$$
$$= 6 \cos 25°$$
$$= 5.44 \text{ teeth/in}$$

We expect to cross fewer teeth per inch in the transverse direction so this makes sense.

The rest of the solution is the same as above.

$$P_t = \frac{N}{d}$$

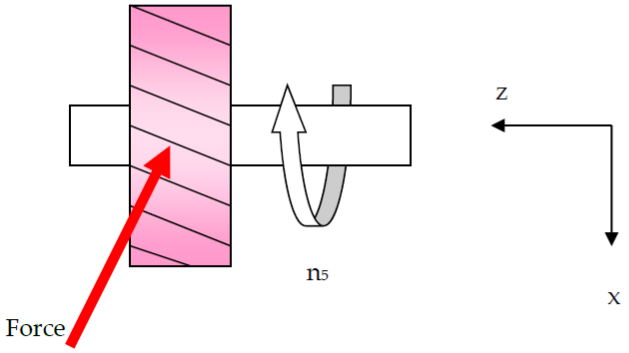
$$d_4 = \frac{N_4}{P_t} = \frac{16 \text{ teeth}}{5.44 \text{ teeth/in}} = 2.94 \text{ inch}$$

$$d_5 = \frac{N_5}{P_t} = \frac{36 \text{ teeth}}{5.44 \text{ teeth/in}} = 6.61 \text{ inch}$$

Shaft-to-shaft distance, y_{45}, $= \frac{d_4 + d_5}{2} = \frac{2.94 + 6.61}{2} = 4.78$ inch

e) Find the direction of the thrust force exerted by gears 4 and 5 on shafts b and c, respectively.

Consider looking at gear 5 and shaft *c* from below. The direction of rotation is as shown. In order to rotate this way, the contact force of gear 4 on gear 5 must be directed as indicated (note: the force is normal to the teeth). This force has a transverse component in the –z direction. Similar arguments can be used for shaft *b*.

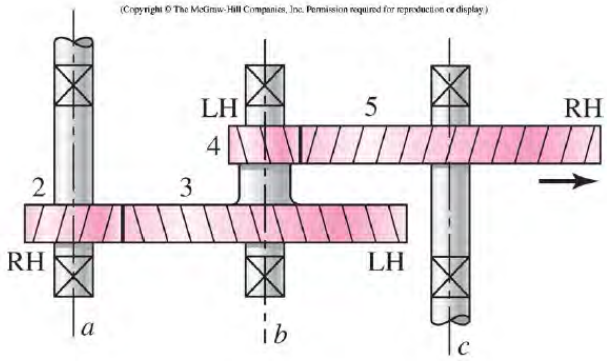


Force on shaft *b* is in +z direction; Force on shaft *c* is in –z direction.

Shigley, Problem 13-47 (8th Ed., 13-33). (To be done by students.)

A gear train is composed of four helical gears with the three shaft axes in a single plane, as shown in the figure. The gears have a normal pressure angle of 20° and a 30° helix angle. Shaft *b* is an idler and the transmitted load acting on gear 3 is 500 lbf. The gears on shaft *b* both have a normal diametral pitch of 7 teeth/in and have 54 and 14 teeth, respectively. Find the forces exerted by gears 3 and 4 on shaft *b* (don't worry about determining moments)

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3. **Shigley, Problem 13-47. (To be done by students.)**

Find the forces exerted by gears 3 and 4 on shaft *b*.

Refer to Fig 13-37 (pg. 704)

The forces from the gears look like:



W_3^t = 500 lb (given)

From Eq. 13-39 (pg. 704)

$$W_r = W\sin\phi_n$$
$$W_t = W\cos\phi_n\cos\psi$$
$$W_a = W\cos\phi_n\sin\psi$$

Where ψ = 30° (given) and φ_n = 20° (given)

Solving for W, W_r, and W_a we find

W_3 = 614.4 lb
W_3^r = 210.2 lb
W_3^a = 288.7 lb

Since gears 3 and 4 are on a common shaft (assumed rotating at constant velocity) the torques must balance. Therefore

$W_3^t d_3 = W_4^t d_4$

$W_4^t = W_3^t d_3 / d_4$
$= W_3^t (N_3 / P_{3t}) / (N_4 / P_{4t})$
$= W_3^t N_3 / N_4$ (since $P_{3t} = P_{4t}$)
$= (500\text{ lb})(54) / (14)$
$= 1929$ lb

Solving for W_r and W_a on gear 4, as above, we find

W_4^r = 811 lb
W_4^a = 1114 lb

The total force on shaft *b* is the sum of the forces from both gears. Referring to the diagram at the start for directions:

$$\vec{W} = \left((210-811)\hat{i} + (-289+1114)\hat{j} + (-500-1929)\hat{k}\right) lb$$
$$= \left(-601\hat{i} + 825\hat{j} - 2429\hat{k}\right) lb$$

Adapted from **Shigley, Problem 14-19**.

A commercial enclosed gear drive consists of a 20° pinion having 16 teeth driving a 48-tooth gear. The pinion speed is 300 rev/min, the face width is 2 in, and the diametral pitch is 6 teeth/in. The gears are grade 1 steel, through-hardened at 200 Brinell, made to No. 6 quality standards, uncrowned, and are to be accurately and rigidly mounted. Assume a pinion life of 10^8 cycles and a reliability of 0.90. The power to be transmitted is 5 hp.

Part 1: To be demonstrated by the TA.

a. Determine the AGMA bending stress in the pinion
b. Determine the corresponding safety factor for the pinion bending stress

Part 2: To be done by students.

c. Determine the AGMA contact stress in the pinion
d. Determine the corresponding safety factor for the pinion contact stress

Solutions

1. Adapted from Shigley, Problem 14-19.

A commercial enclosed gear drive is to be designed. The drive is to consist of two spur gears with a 20° pressure angle. The pinion is to have 16 teeth driving a 48- tooth gear. The pinion speed is 300 rev/min, the face width is 2 in, and the diametral pitch is 6 teeth/in. The gears are Grade 1 steel, through-hardened at 200 Brinell, made to No. 6 quality standards. The gears will be uncrowned, centered on their shafts between bearings, and are to be accurately and rigidly mounted. Assume a pinion life of 10^8 cycles and a reliability of 0.90. The power to be transmitted is 5 hp.

Part 1: To be demonstrated by the TA.

a. Determine the AGMA bending stress in the pinion

Working from the "roadmap" in Fig 14-17 (p. 772),

$$d_p = \frac{N_p}{P_d} = \frac{16 \text{ teeth}}{6 \text{ teeth/in}} = 2.667 \text{ in}$$ (pinion pitch diameter)

$$V = \frac{\pi d_p n_p}{12} = \frac{\pi(2.667)(300)}{12} = 209.5 \text{ ft/min}$$ (pitch line velocity, note the units)

$$W^t = 33\,000\frac{H}{V} = 33\,000\frac{5}{209.5} = 787.6 \text{ lbf}$$ (transmitted load)

the gear bending stress equation is given by

$$\sigma = W^t K_o K_v K_s \frac{P_d}{F}\frac{K_m K_b}{J}$$ Equation 14-15, page 751

We know W^t from above, we are given P_d = 6 teeth/in, and we are given the face width, F = 2in. To compute σ, we need to solve for all of the coefficients.

K_o: overload factor

Since we are given the fact that the driver and driven machines are smooth. We assume the driven machine have uniform loading. From the table on the bottom of p. 772,

$K_o = 1$

K_v: velocity factor

Although we could use Figure 14-9 (p. 764) to find K_v, it would be difficult to get an accurate value due to the steepness of the curve and low resolution of the V values on the x-axis.

Instead, use the expressions in eqns. 14-27 and 14-28 on p. 763.

With V = 209.5 ft/min (found above) and Q_v = 6 (given in question) we have

Eq. (14-28): $B = 0.25(12 - Q_v)^{2/3} = 0.25(12 - 6)^{2/3} = 0.8255$

$A = 50 + 56(1 - B) = 50 + 56(1 - 0.8255) = 59.77$

Eq. (14-27): $$K_v = \left(\frac{A + \sqrt{V}}{A}\right)^B = \left(\frac{59.77 + \sqrt{209.5}}{59.77}\right)^{0.8255} = 1.196$$

K_v = 1.20

K_s: size factor

Normally, the size factor reflects non-uniformity of material properties due to size. It reflects the Marin factor for size in the calculation of endurance strength. The idea is that the larger the volume of the material, the higher the probability of a defect (stress raiser) that reduces the fatigue life.

Mott gives a table with suggested size factors. In this case, K = 1.00 for diametrical pitch = 6 teeth/inch

TABLE 9–2 Suggested Size Factors, K_s

Diametral pitch, P_d	Metric module, m	Size factor, K_s
≥ 5	≤ 5	1.00
4	6	1.05
3	8	1.15
2	12	1.25
1.25	20	1.40

Using the Shigley approach here we get:

From eq. (a) on p. 751

From Eq. (*a*), Sec. 14-10, with F = 2 in

$$(K_s)_P = 1.192\left(\frac{F\sqrt{Y}}{P}\right)^{0.0535} = 1.192\left(\frac{2\sqrt{0.296}}{6}\right)^{0.0535} = 1.088$$

F = 2in (given), the Lewis factor which governs tooth size is Y = 0.296 for the pinion (see Table 14-2, p. 744), P_d = 6 (given)

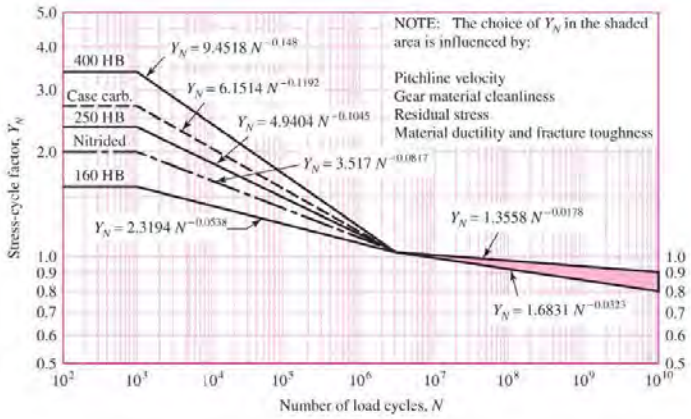
As stated in the Shigley text (page 765), "You may set K_s = 1.0 or you may elect to

S_t: bending strength (Fig. 14-2, p. 752)

$$S_t = 77.3H_B + 12{,}800 \text{ psi}$$
$$= 77.3(200) + 12{,}800 \text{ psi} \qquad (H_B = 200 \text{ is the Brinell hardness})$$
$$= 28.26 \text{ ksi}$$

Y_N: repeated bending stress cycle factor

With 10^8 cycles, $Y_N \approx 0.95$ (read from Fig 14-14, pg. 755)
(Shigley uses a less conservative value)



Using the equation in Figure 14-14 we get for the pinion, $N = 10^8$ cycles,:

$$(Y_N)_P = 1.3558(10^8)^{-0.0178} = 0.977$$

K_T: Temperature factor

Since no reason to expect T > 250°F, $K_T = 1$ (Section 14-15, p. 770)

K_R: Reliability factor

With reliability of 0.90, $K_R = 0.85$ (from Table 14-10, p. 770)

Combining all terms,

Eq. (14-41): $$(S_F)_P = \left[\frac{S_t Y_N / (K_T K_R)}{\sigma}\right] = \frac{28\,260(0.977) / [(1)(0.85)]}{13\,904} = 2.34 \quad Ans$$

The safety factor for the pinion in bending is 2.34.

Part 2: To be done by students.

c. Determine the AGMA contact stress in the pinion

The process is the same as above. The required equation is eq. 14-16 (p. 752)

$$\sigma_c = C_P \sqrt{W^t K_o K_v K_s \frac{K_m \; C_f}{d_p F \;\; I}}$$

The terms W^t = 787.6 lb, K_o = 1, K_v = 1.196, K_s = 1.088, K_m = 1.223, d_p = 2.667″, and F = 2″ were either given or solved for in the previous part of the problem. It remains to determine C_P, C_f, and I.

C_P: Elastic coefficient

C_P = 2300 $psi^{1/2}$ (from steel pinion-steel gear in Table 14-8, p. 762)

C_f: Surface condition coefficient is only applied when a detrimental surface finish is known to exist. Otherwise just us 1.0

C_f = 1 (as noted on p. 764)

I: Surface-Strength Geometry Factor

This factor is also known as the Pitting Resistance Factor. It is related to the Hertz contact stress equations.

Eq. (14-22): $m_G = N_G/N_P = 48/16 = 3$

As noted below eq. 14-23 (p. 760) the load sharing ratio m_N = 1 for spur gears.

For a spur gear, φ_t is just φ, the pressure angle. As given in the problem description, φ = 20° in this case.

Substituting to the expression for I above, we get

Eq. (14-23): $$I = \frac{\cos\phi_t \sin\phi_t}{2(m_N)} \frac{m_G}{m_G + 1} = \frac{\cos 20^\circ \sin 20^\circ}{2(1)} \left(\frac{3}{3+1} \right) = 0.1205$$

Insert all of the coefficients into the contact stress expression

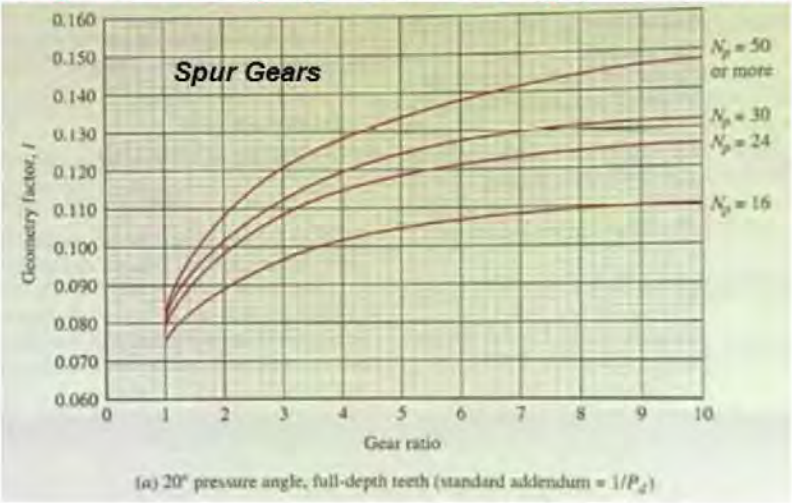
Eq. (14-16):

$$(\sigma_c)_P = C_p \left(W^t K_o K_v K_s \frac{K_m}{d_P F} \frac{C_f}{I} \right)_P^{1/2}$$

$$= 2300 \left[787.8(1)(1.196)(1.088) \left(\frac{1.2207}{2.667(2)} \right) \left(\frac{1}{0.1205} \right) \right]^{1/2} = 101\,485 \text{ psi} \quad Ans.$$

The contact stress is 101.5 ksi

Note: The Mott text has the Geometry Factor I for Spur Gears. Here that values are slightly different but easier to determine. From the Figure 9-17 (Mott page 388) we get I = 0.097 for $m_G = 3$. It wil change the solution slightly. We will use the Shigley solution for the next calculations.



(a) 20° pressure angle, full-depth teeth (standard addendum = $1/P_d$)

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d. Determine the corresponding safety factor for the pinion in terms of contact stress

The safety factor is given by eq. 14-42 (p. 771)

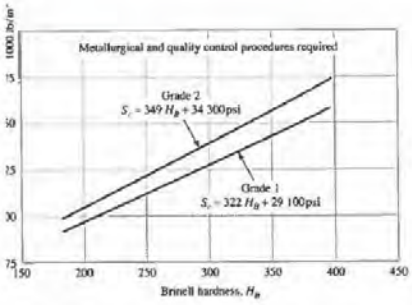
$$S_H = \frac{S_C}{\sigma_C}\frac{Z_N C_H}{K_T K_R}$$

The same process as before is used. In part a., it was already determined that $K_T = 1$ and $K_R = 0.85$. For σ_c we use our result from part c., $\sigma_c = 101.5$ ksi.

S_c: Contact-fatigue strength

$$S_c = 322H_B + 29{,}100 \text{ psi}$$
$$= 322(200) + 29{,}100$$
$$= 93{,}500 \text{ psi}$$

(Fig 14-5, p. 755)



Z_N: Stress Cycle Factor

From Figure 14-15 (p. 769) with 10^8 cycles, one can choose the centre of the shaded region (more conservative than Shigley)

$Z_N = 0.90$

Using the curve fit equation we get: $(Z_N)_P = 1.4488(10^8)^{-0.023} = 0.948$

C_H: hardness ratio factor

As noted in Sec 14-12 (p. 753), C_H applies to the gear only. We can either neglect the term or use $C_H = 1$, with the same effect.

$C_H = 1$

Eq. (14-42) with $Z_N = 0.948$:

$$(S_H)_P = \left[\frac{S_c Z_N / (K_T K_R)}{\sigma_c}\right]_P = \frac{93\,500(0.948) / [1(0.85)]}{101\,485} = 1.03 \quad Ans.$$

With Z = 0.9, we get the final expression as

$$S_H = \frac{(93.5\text{ ksi})(0.90)(1)}{(1)(0.85)} \frac{1}{101.5\text{ ksi}} \approx 0.97$$

$S_H = 0.97 - 1.02$

With a safety factor less than or barely above unity, pinion wear is a concern so hardness should be increased. (It won't disintegrate immediately, but it will not last 10^8 cycles with 0.90 reliability.)

Another option is to increase the face width to say 2.5 or 3 inches. Here the Load Distribution Factor K_m will change very slightly. But the contact stress will be reduced to the point where the safety may be more acceptable.

1. Adapted from Shigley, Problem 14-22.

A commercial enclosed gear drive is to be designed. The drive is to consist of two helical gears with a 20° pressure angle and a helix angle of 25°. The pinion is to have 16 teeth driving a 48- tooth gear. The pinion speed is 300 rev/min, the face width is 2 in, and a normal diametrical pitch of 6 teeth/in. The gears are Grade 1 steel made to No. 6 quality standards. The gears will be uncrowned, centered on their shafts between bearings, and are to be accurately and rigidly mounted. Assume a pinion life of 10^8 cycles and a reliability of 0.90. The power to be transmitted is 5 hp.

Part 1: To be demonstrated by the TA.

Note: I moved form 30° helix angle to 25° helix in order to use the charts in Mott text. The Shigley solutions are for ψ=30°

a. Determine the AGMA bending stress in the pinion

Working from the "roadmap" in Fig 14-17 (p. 772),

$P_t = P_n \cos\psi = 6\cos 25^\circ = 5.438$ teeth/in (Eq. 13-16 pinion transverse pitch)

$$d_P = \frac{16}{5.438} = 2.942 \text{ in}, \quad d_G = \frac{48}{16}(2.942) = 8.827 \text{ in}$$

Eq. (13-34): $$V = \frac{\pi d_P n_P}{12} = \frac{\pi(2.942)(300)}{12} = 231.1 \text{ ft/min}$$

Eq. (13-35): $$W^t = 33\,000\frac{H}{V} = 33\,000\frac{5}{231.1} = 714.0 \text{ lbf}$$

$$\phi_t = \tan^{-1}\left(\frac{\tan 20^\circ}{\cos 25^\circ}\right) = 21.9^\circ$$

Note : Helical gear slightly larger diameter (2.942">2.667" for spur gear) and therefore slightly higher velocity and lower transverse force (714 lbs <787.6lbf for spur gear).

the helical gear bending stress equation is the same as the spur gear equation:

$$\sigma = W^t K_o K_v K_s \frac{P_d}{F}\frac{K_m K_b}{J}$$

Equation 14-15, page 751

We know W^t from above, we are given P_n = 6 teeth/in, and we are given the face width, F = 2in. To compute σ, we need to solve for all of the coefficients.

K_o: overload factor (same as Problem 14-20)

As in Problem 14-20, this does not change

$K_o = 1$

K_v: velocity factor

Although we could use Figure 14-9 (p. 764) to find K_v, it would be difficult to get an accurate value due to the steepness of the curve and low resolution of the V values on the x-axis.

Instead, use the expressions in eqns. 14-27 and 14-28 on p. 763.

With V = 231.1 ft/min (found above) and $Q_v = 6$ (given in question) we have

Eq. (14-28): $B = 0.25(12 - Q_v) = 0.25(12 - 6)^{2/3} = 0.8255$

$A = 50 + 56(1 - B) = 50 + 56(1 - 0.8255) = 59.77$

Eq. (14-27): $K_v = \left(\frac{A + \sqrt{V}}{A}\right)^B = \left(\frac{59.77 + \sqrt{241.8}}{59.77}\right)^{0.8255} = 1.210$ 1.206

231.1

We can use $K_v = 1.21$

K_s: size factor (Same as 14-20)

Normally, the size factor reflects non-uniformity of material properties due to size. It reflects the Marin factor for size in the calculation of endurance strength. The idea is that the larger the volume of the material, the higher the probability of a defect (stress raiser) that reduces the fatigue life.

Mott gives a table with suggested size factors. In this case, K = 1.00 for diametrical pitch >5 teeth/inch

TABLE 9–2 Suggested Size Factors, K_s

Diametral pitch, P_d	Metric module, m	Size factor, K_s
≥ 5	≤ 5	1.00
4	6	1.05
3	8	1.15
2	12	1.25
1.25	20	1.40

If we continue to use the Shigley approach here we can continue to use P_n =6 to get:

From eq. (a) on p. 765

From Eq. (*a*), Sec. 14-10, with *F* = 2 in

$$(K_s)_P = 1.192\left(\frac{F\sqrt{Y}}{P}\right)^{0.0535} = 1.192\left(\frac{2\sqrt{0.296}}{6}\right)^{0.0535} = 1.088$$

F = 2in (given), the Lewis factor which governs tooth size is Y = 0.296 for the pinion (see Table 14-2, p. 744), P_d = 6 (given)

As stated in the Shigley text (page 765), "You may set K_s = 1.0 or you may elect to use the preceding equation (above). This is a point to discuss with the instructor." I normally will suggest we default to 1.0.

However, for this problem, we shall proceed with the Shigley value derived above.

K_s = 1.09 (same as Problem 14-20)

K_m: Load-distribution factor

The load distribution factor modifies the stress equation to reflect nonuniform distribution of load across the line of contact. See section 14-11

From eq. 14-30 (p. 765)

$$K_m = 1 + C_{mc}\left(C_{pf}C_{pm} + C_{ma}C_e\right)$$

The "C" terms are all given on pp. 765-7.

Eq. (14-31): C_{mc} = 1 (given that we have uncrowned teeth)

The C_{pf} value defines the face length. The expectation is that the wider the tooth, the harder it will be to maintain uniform distribution.

Eq. (14-32) for 1 < F < 17inches:

0.0554

$$C_{pf} = \frac{F}{10d_p} - 0.0375 + 0.0125F = \frac{2}{10(2.667)} - 0.0375 + 0.0125(2) = 0.0625$$

2.942

Eq.(4-33) assuming gears are centered between bearings: C_{pm} = 1

For the *Mesh Alignment Factor* C_{ma} we use the Figure 14-11 (page 767)

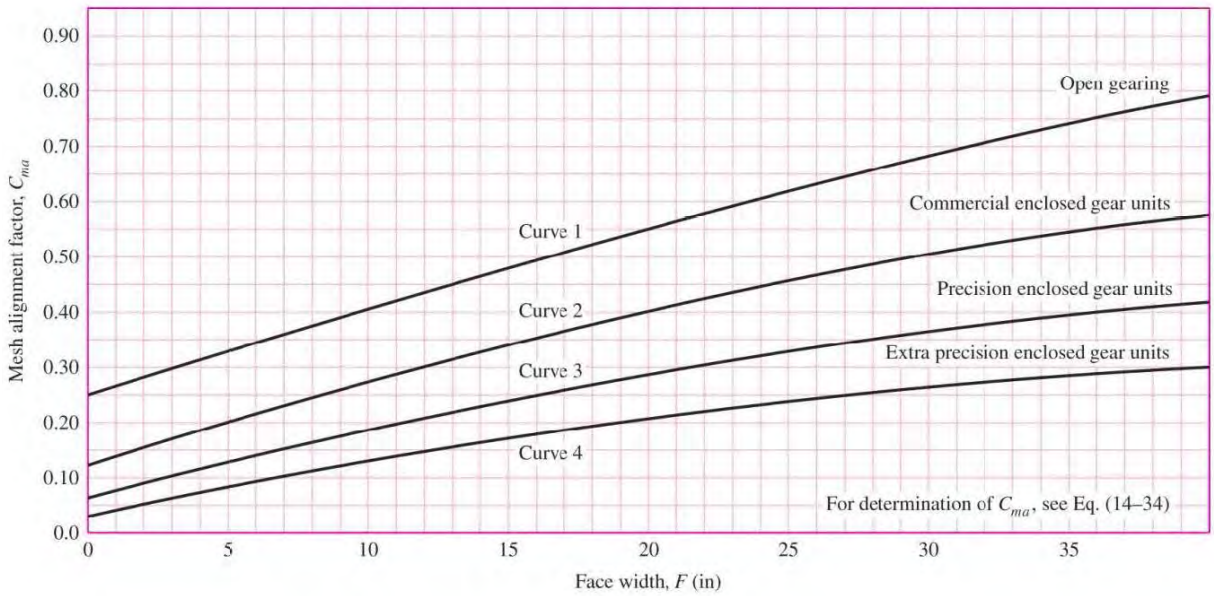


Fig. 14-11: C_{ma} = 0.16 (Face width = 2.0" and commercial enclosed gear unit (stated in problem)

Eq. (14-35): C_e = 1 (eq. 14-35, no information to suggest it should be 0.8)

Putting it all together:

0.0554 1.215

Eq. (14-30): $K_m = 1 + C_{mc}\left(C_{pf}C_{pm} + C_{ma}C_e\right) = 1 + 1\left[0.0625(1) + 0.16(1)\right] = 1.2225$

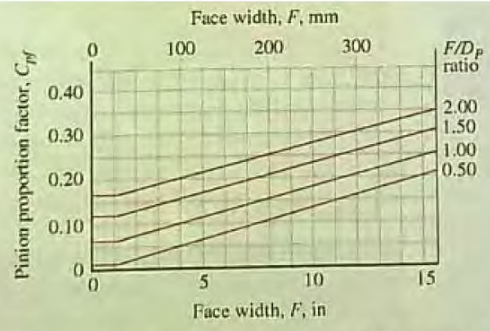
K_m = 1.22 (basically the same as Problem 14-20)

Note: the procedure in the Mott text is simplified uses only the two main C factors

$$K_m = 1.0 + C_{pf} + C_{ma} \qquad (9\text{–}17)$$

where C_{pf} = pinion proportion factor (see Figure 9–12)

C_{ma} = mesh alignement factor (see Figure 9–13)



D_P = Pinion diameter

For $F/D_P < 0.50$, use curve for $F/D_P = 0.50$

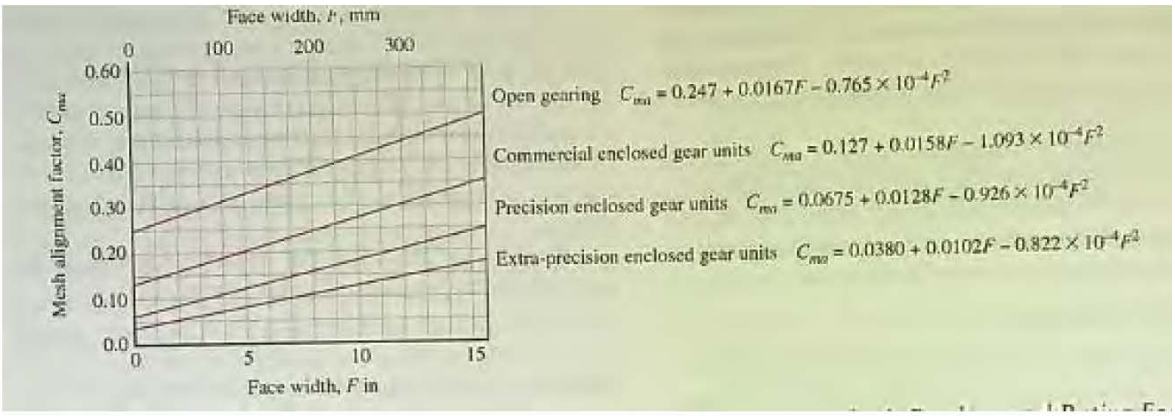
When $F \le 1.0$ in. ($F \le 25$ mm)

$$C_{pf} = \frac{F}{10D_P} - 0.025$$

When $1.0 \le F < 15$,

$$C_{pf} = \frac{F}{10D_P} - 0.0375 + 0.0125F$$

FIGURE 9–12 Pinion proportion factor, C_{pf} (Extracted from AGMA 2001 Standard, *Fundamental Rating Factors and Calculation Methods for Involute Spur and Helical Gear Teeth*, with permission of the publisher, American Gear Manufacturers Association, 1001 North Fairfax Street, 5th floor, Alexandria, VA 22314)



Here we get C_{pf} = 0.0554 (same as above) and C_{ma} = 0.16 (same as above)
So K_m = 1 + 0.0554 + 0.16 = 1.22 (as above).

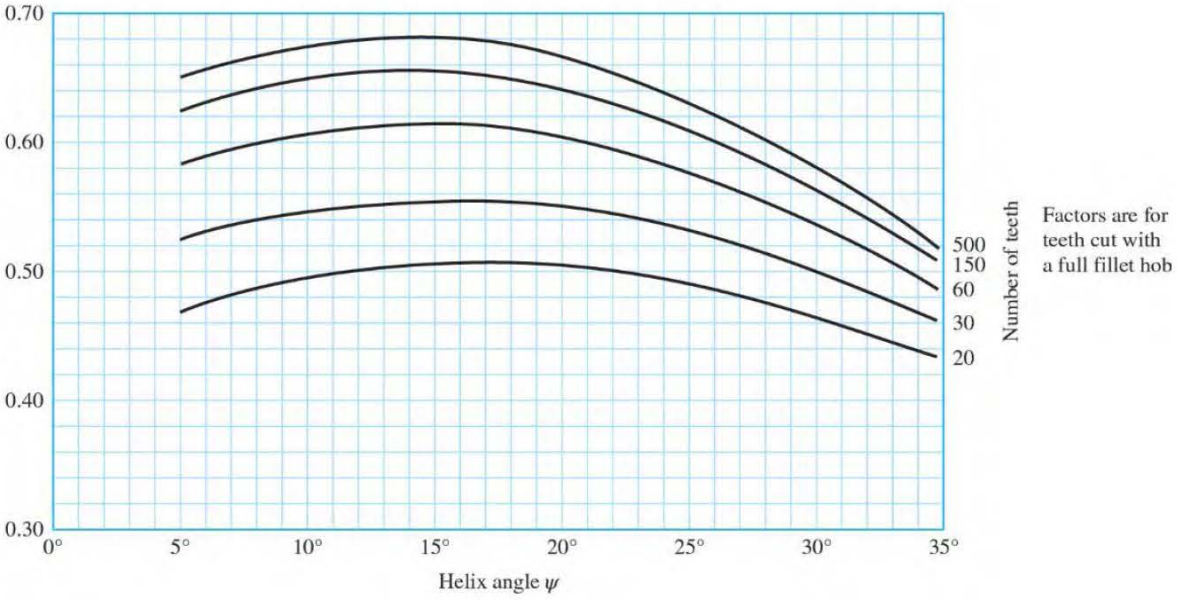
K_b: rim thickness factor

From eq. 14-40 and Fig. 14-16 (p. 770), no reason to believe m_b < 1.2, so use

K_b = 1

J: Bending-Strength Geometry Factor

For Helix Gears we need to use the Figures 14-7 and 14-8 (pages 760/1). As you will see from the figure below, the lowest pinion gear is 20 teeth. Our question asks for N_p = 16. We can do a little interpolation to get the Geometry Factor J' at the helix angle, ψ, of 25°. For 20 teeth at ψ= 25° we read 0.49 If we extrapolate a bit below for 16 teeth we can estimate J' as 0.47. This value is likely a little conservative.



Next we need to need to modify the value of J' with the modification factor from Figure 14-8. This factor is based on the fact that the gear will have number of teeth different than the 75 tooth gear that is the standard used by AGMA. I use the term $m_{J'}$ as the variable for this value.

manner to Problem 14-20. However in this case, we want to determine a material strength number to give a realistic Safety factor of between 1.0 and 1.5. Hence we solve for S_t with S_F = 1.0. This is the minimum strength number required for the pinion.

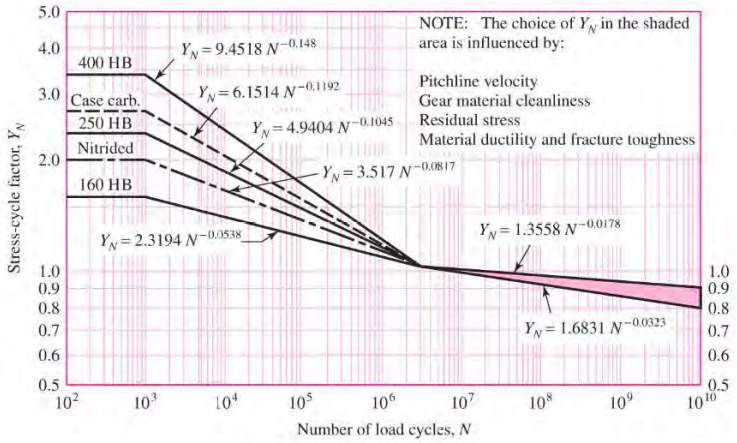
The gear strength number is related to the Brinell Hardness, HB.

S_t: bending strength (Fig. 14-2, p. 752) for Grade 1 gears

$$S_t = 77.3H_B + 12{,}800 \text{ psi}$$

Y_N: repeated bending stress cycle factor (same as Problem 14-20)

With 10^8 cycles, $Y_N \approx 0.95$ (read from Fig 14-14, pg. 755)
(Shigley uses a less conservative value)



Using the equation in Figure 14-14 we get for the pinion, $N = 10^8$ cycles,:

$$(Y_N)_P = 1.3558(10^8)^{-0.0178} = 0.977$$

The same as Problem 14-20.

K_T: Temperature factor (same as 14-20)

Since no reason to expect T > 250°F, K_T = 1 (Section 14-15, p. 770)

K_R: Reliability factor (same as 14-20)

With reliability of 0.90, K_R = 0.85 (from Table 14-10, p. 770)

Combining all terms,

Eq. (14-41): $$(S_F)_P = \left[\frac{S_t Y_N / (K_T K_R)}{\sigma}\right] = \frac{28\,260(0.977) / [(1)(0.85)]}{13\,904} = 2.34 \quad Ans$$

The minimum bending strength number S_t = 5947psi = 5.9 ksi.

c. Determine the safety factor for bending if a 200HB Grade 1 through-hardened gear is selected

As our required strength number is part (b) is very low, we can expect a high safety factor.

S_t: bending strength (Fig. 14-2, p. 752) for Grade 1 gears

$$S_t = 77.3H_B + 12{,}800 \text{ psi}$$

For H_B = 200, we get S_t =77.3(200) + 12800 = 28260psi = 28.3ksi

Safety Factor $(S_F)_{Pinion}$ = 28.3/5.9 = 4.8.

In reality, the lowest value of H_B for the pinion can be used with high confidence.

d. Determine the AGMA contact stress in the pinion

The process is the same as above. The required equation is eq. 14-16 (p. 752)

$$\sigma_c = C_P \sqrt{W^t K_o K_v K_s \frac{K_m C_f}{d_p F I}}$$

The terms W^t = 714 lb, K_o = 1, K_v = 1.21, K_s = 1.09, K_m = 1.22, d_p = 2.942", and F = 2" were either given or solved for in the previous part of the problem. The values for C_P, and C_f, are the also the same as Problem 14-20. However the surface strength geometry factor, I will be different for the helical gear.

C_P: Elastic coefficient (Same as Problem 14-20)

C_P = 2300 $psi^{1/2}$ (from steel pinion-steel gear in Table 14-8, p. 762)

C_f: Surface condition coefficient is only applied when a detrimental surface finish is known to exist. Otherwise just us 1.0 (Same as Problem 14-20)

C_f = 1 (as noted on p. 764)

I: Surface-Strength Geometry Factor

This factor is also known as the Pitting Resistance Factor. It is related to the Hertz contact stress equations. We can use the equations in Shigley or Table 10-1 from Mott.

21.9

Eq. (14-23): $$I = \frac{\cos\phi_t \sin\phi_t}{2m_N} + \frac{m_G}{m_G + 1} = \left[\frac{\sin 22.8^\circ \cos 22.8^\circ}{2(0.6937)}\right]\left(\frac{3}{3+1}\right) = 0.193$$

0.190

This value of I is lower than that obtained from Mott. It is a little more conservative and so we will proceed with this value. However, the Mott is faster and suitable for most applications.

Insert all of the coefficients into the contact stress expression

Pinion tooth wear

Eq. (14-16):

$$(\sigma_c)_P = C_p\left(W^t K_o K_v K_s \frac{K_m}{d_P F}\frac{C_f}{I}\right)_P^{1/2}$$

1.22

1.22

$$= 2300\left\{682.3(1)(1.21)(1.088)\left[\frac{1.213}{3.079(2)}\right]\left(\frac{1}{0.193}\right)\right\}^{1/2} = 69\,640 \text{ psi} \quad \textit{Ans.}$$

714 2.942 0.19 73967

Note: If Mott value of I is used, contact stress number is 71.0ksi versus 74.0 ksi.

e. Determine the minimum strength number for the pinion in contact

To find the safety factor in pinion contact stress, use the allowable stress in eq. 14-42 (p. 771)

$$(S_H)_P = \left[\frac{S_c Z_N / (K_T K_R)}{\sigma_c}\right]_P$$

The σ_c is the stress calculated in part d. The other values are found in a similar manner to Problem 14-20. However in this case, we want to determine a material strength number to give a realistic Safety factor of between 1.0 and 1.5. Hence we solve for S_H with S_F = 1.0. This is the minimum strength number required for the pinion.

The gear strength number is related to the Brinell Hardness, HB.

S_H: contact strength number (Fig. 14-5, p. 755) for Grade 1 gears

$$S_H = 322H_B + 29{,}100 \text{ psi}$$

Z_N: repeated contact stress cycle factor (same as Problem 14-20)

With 10^8 cycles, $Z_N \approx 0.95$ (read from Fig 14-14, pg. 755)
(Shigley uses a less conservative value)

(Eq. 14-43):

$$(S_H)_P = \left[\frac{S_c Z_N / (K_T K_R)}{\sigma_c}\right]_P = \frac{93\,500(0.948) / [(1)(0.85)]}{69\,640} = 1.50 \quad Ans.$$

S_c 0.95 1.0 73967

The minimum contact strength number S_c = 66181psi = 66.2 ksi.

f. Determine the safety factor for contact stress if a 200HB Grade 1 through-hardened gear is selected

S_H: contact strength number (Fig. 14-5, p. 755) for Grade 1 gears

$$S_H = 322H_B + 29{,}100 \text{ psi}$$

For H_B = 200, we get S_H =322(200) + 29100 = 93500psi = 93.5ksi

Safety Factor $(S_F)_{Pinion}$ = 93.5/66.2 = 1.41

Again, it is contact stress that dominates this design. A HB of 200 is very suitable for this gear pinion.

g. Select a suitable steel alloy and heat treatment to meet the requirement (HB200). What is the final safety factor.

Suitable steel alloys can be found using Table 9-8 from Mott. For through-hardened gears, one can select the 41XX Chrome molybdenum steels or the 43XX Nickel chrome molybdenum steels.

Let select the 4140 steel (0.4% Carbon content)

TABLE 9–8 Examples of Gear Materials

Heat treatment	Typical alloys (SAE numbers)
Through-hardened or Case-hardened by flame or induction hardening	1045, 4140, 4150, 4340, 4350
Carburizing, case-hardened	1020, 4118, 4320, 4820, 8620, 9310

From Appendix A3

4130	WQT 400	234	1610	197	1360	12	401
4140[1]	Annealed	95	655	54	372	26	197
4140	OQT 1300	117	807	100	690	23	235
4140	OQT 1000	168	1160	152	1050	17	341
4140	OQT 700	231	1590	212	1460	13	461
4140	OQT 400	290	2000	251	1730	11	578
4150	Annealed	106	731	55	379	20	197
			880	116	800	20	262

We require a minimum of HB200. The annealed version of the metal is very close HB=197 but it does meet the minimum requirement. Hence we shall select 4140 OQT 1300 (oil quenched and tempered at 1300°F). Here HB=235.

So for H_B = 235, we get S_H =322(235) + 29100 = 104770psi = 104.8ksi

Safety Factor $(S_F)_{Pinion}$ = 104.8/66.2 = 1.58 $(S_F)^2$ = 2.50

If we remove the requirement of HB 200 and just stick to the requirement of Safety factor between 1.0 and 1.5 we can select the annealed 4140 steel alloy.

With H_B = 197, we get S_H =322(197) + 29100 = 92534psi = 92.5ksi

Safety Factor $(S_F)_{Pinion}$ = 92.5/66.2 = 1.40 $(S_F)^2$ = 1.96

Both the 4140 annealed and OQT1300 versions of the alloy will be suitable for this application.

From Shigley Appendix A-21, we can select the 4140 Annealed (HB=197) or the 4140 Q&T 650°C (1200°F) with HB = 230

The safety factor for the HB = 230 is 1.56. $(S_F)^2$ = 2.50

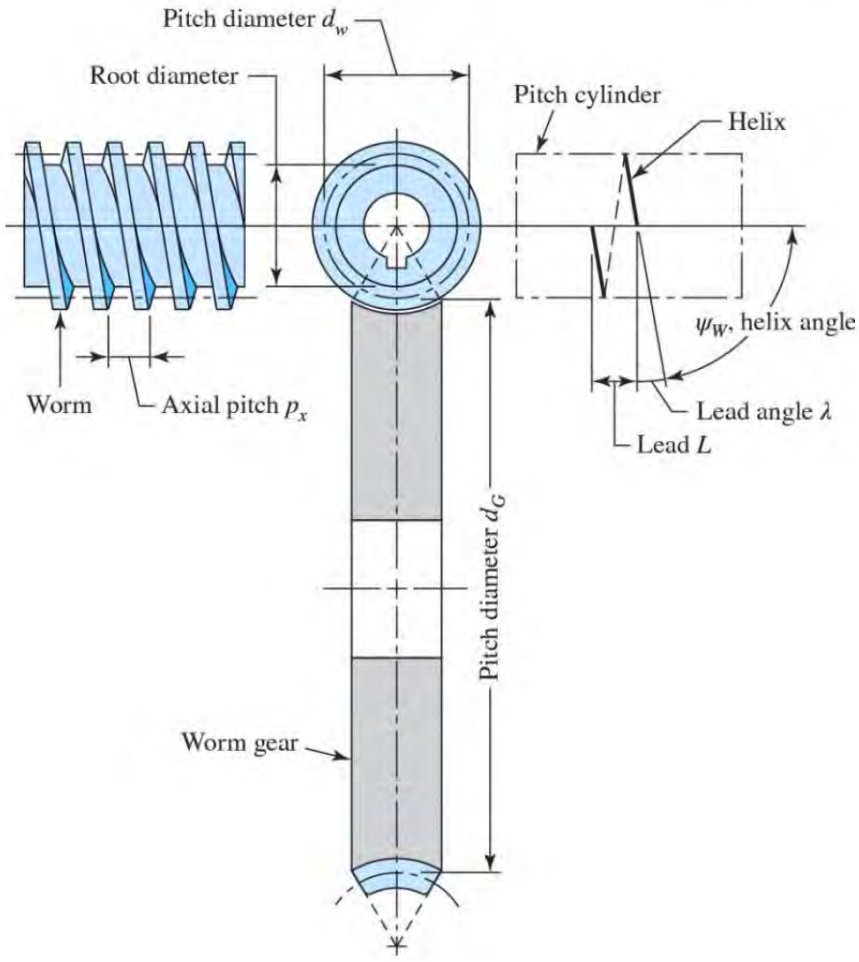
If we are to compare the safety factors between contact and bending based on tangential load, W_t, we must square the SF as done above. In this case each of the safety factors for contact are below the safety factor for bending from part c), Safety Factor $(S_F)_{Pinion}$ = 28.3/5.9 = 4.8.

1. Adapted from Shigley, Problem 14-22.

A single-thread steel worm gear rotates at 1725rpm meshing with a 56-tooth worm gear transmitting 1 HP to the output shaft. The pitch diameter of the worm is 1.5 inches. The tangential diametrical pitch of the gear is 8 teeth/inch and the normal pressure angle is 20°. The ambient temperature is 70°F and the service (application) factor is $K_a = 1.25$, the design factor, $n_d = 1.0$, the gear face is F = 0.5 inches, the lateral case area is 850 in^2 per side of gear box and the gear is made from sand-cast phosphorous bronze.

Part 1: To be demonstrated by the TA.

a. Determine and evaluate the geometric properties of the gears.



Use Table 15-8 (page 815) to assist with the standard geometric properties
Note: the pitch diameter for the worm d_p is the same as the mean diameter for the worm, d_m. This diameter is normally just represented as the worm diameter d_w(D_w in the Mott text).

$N_W = 1$, $N_G = 56$, $P_t = 8$ teeth/in, $d_w = 1.5$ in, $H_o = 1$hp, $\phi_n = 20°$, $t_a = 70°$F,

$K_a = 1.25$, $n_d = 1$, $F_e = 0.5$ in, $A = 850$ in^2

(a) $m_G = N_G/N_W = 56$, $d_G = N_G/P_t = 56/8 = 7.0$ in
$p_x = \pi/8 = 0.3927$ in, $C = (1.5 + 7)/2 = 4.25$ in

Quantity	Symbol	ϕ_n 14.5° $N_W \le 2$	ϕ_n 20° $N_W \le 2$	ϕ_n 25° $N_W > 2$
Addendum	a	$0.3183p_x$	$0.3183p_x$	$0.286p_x$
Dedendum	b	$0.3683p_x$	$0.3683p_x$	$0.349p_x$
Whole depth	h_t	$0.6866p_x$	$0.6866p_x$	$0.635p_x$

*The table entries are for a tangential diametral pitch of the gear of $P_t = 1$.

Table 15–8
Worm-Gear Geometry

$$a = \frac{p_x}{\pi} = 0.3183\,p_x \qquad \textbf{(15-39)}$$

$$b = \frac{1.157\,p_x}{\pi} = 0.3683\,p_x \qquad \textbf{(15-40)}$$

$$h_t = \begin{cases} \dfrac{2.157\,p_x}{\pi} = 0.6866\,p_x & p_x \ge 0.16 \text{ in} \\ \dfrac{2.200\,p_x}{\pi} + 0.002 = 0.7003\,p_x + 0.002 & p_x < 0.16 \text{ in} \end{cases} \qquad \textbf{(15-41)}$$

$$d_o = d + 2a \qquad \textbf{(15-42)}$$

$$d_r = d - 2b \qquad \textbf{(15-43)}$$

$$D_t = D + 2a \qquad \textbf{(15-44)}$$

$$D_r = D - 2b \qquad \textbf{(15-45)}$$

$$c = b - a \qquad \textbf{(15-46)}$$

Eq. (15-39): addendum $a = p_x/\pi = 0.3927/\pi = 0.125$ in

Eq. (15-40): dedendum $b = 0.3683\,p_x = 0.1446$ in

Eq. (15-41): whole depth $h_t = 0.6866\,p_x = 0.2696$ in

Eq. (15-42): worm outside diameter $d_o = 1.5 + 2(0.125) = 1.75$ in

Eq. (15-43): worm root diameter $d_r = 1.5 - 2(0.1446) = 1.21$ in

Eq. (15-44): worm-gear throat diameter $D_t = 7 + 2(0.125) = 7.25$ in

Eq. (15-45): worm-gear pitch diameter $D_r = 7 - 2(0.1446) = 6.711$ in

Eq. (15-46): clearance $c = 0.1446 - 0.125 = 0.0196$ in

Eq. (15-47): maximum worm face width $(F_W)_{\max} = 2\sqrt{2(7)(0.125)} = 2.646$ in

Recall V= π d n/12

$$V_W = \pi(1.5)(1725/12) = 677.4 \text{ ft/min}$$

$$V_G = \frac{\pi(7)(1725/56)}{12} = 56.45 \text{ ft/min}$$

Recommended range for worm pitch diameter as a function of center distance C,

$$\frac{C^{0.875}}{3.0} \le d_W \le \frac{C^{0.875}}{1.7} \qquad \textbf{(13-26)}$$

Relation between *lead* L and *lead angle* λ,

$$L = p_x N_W \qquad \textbf{(13-27)}$$

$$\tan\lambda = \frac{L}{\pi d_W} \qquad \textbf{(13-28)}$$

Eq. (13-27): Lead $L = p_x N_W = 0.3927$ in

Eq. (13-28): lead angle $\lambda = \tan^{-1}\left(\frac{0.3927}{\pi(1.5)}\right) = 4.764°$

$$P_n = \frac{P_t}{\cos\lambda} = \frac{8}{\cos 4.764°} = 8.028$$

$$p_n = \frac{\pi}{P_n} = 0.3913 \text{ in}$$

Sliding Velocity and Torque

$$V_s = \frac{\pi n_W d_m}{12\cos\lambda} \qquad \textbf{(15-30)}$$

$$T_G = \frac{W^t D_m}{2} \qquad \textbf{(15-31)}$$

Eq. (15-30/15-62): $V_s = \frac{\pi(1.5)(1725)}{12\cos 4.764°} = 679.8$ ft/min

With C = (1.5+7)/2 = 4.25 inches. From equation 13-26 we see that 1.182 < d < 2.09. Our d = 1.50" so we are within the recommended worm diameter for this mesh. We will now perform the analysis to see if the AGMA wear condition is exceeded.

With a lead angle of 4.764° < 5°, we can expect this worm gear to be "self-locking." The gear will not drive the worm, normally referred to as "back-driving."

Part 2: To be done by the student.

b. Determine the transmitted gear forces and the mesh efficiency.

To find the coefficient of sliding friction, one can use Figure 10-25 from Mott or the curve fit equation from AGMA for sliding velocities greater than 10ft/min (Equation 15-38 in Shigley (Equation 10-27 in Mott)

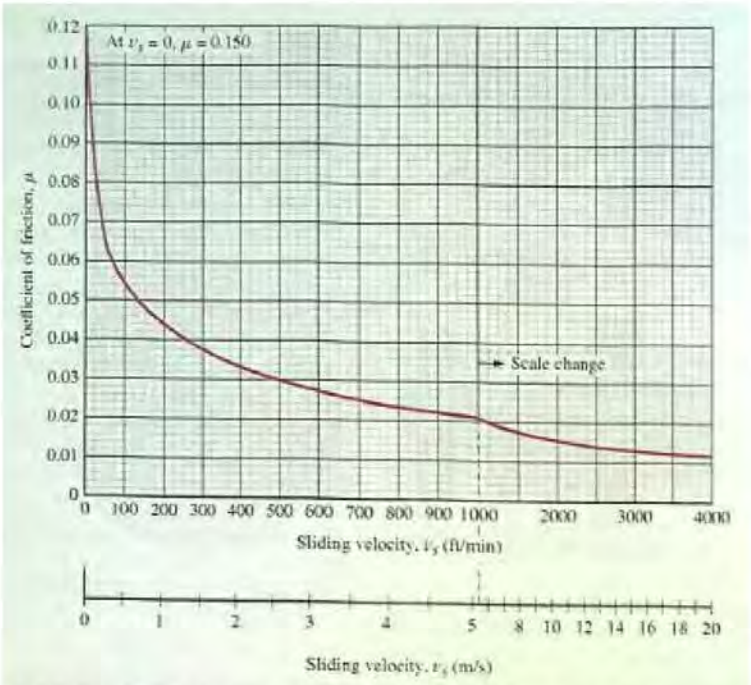


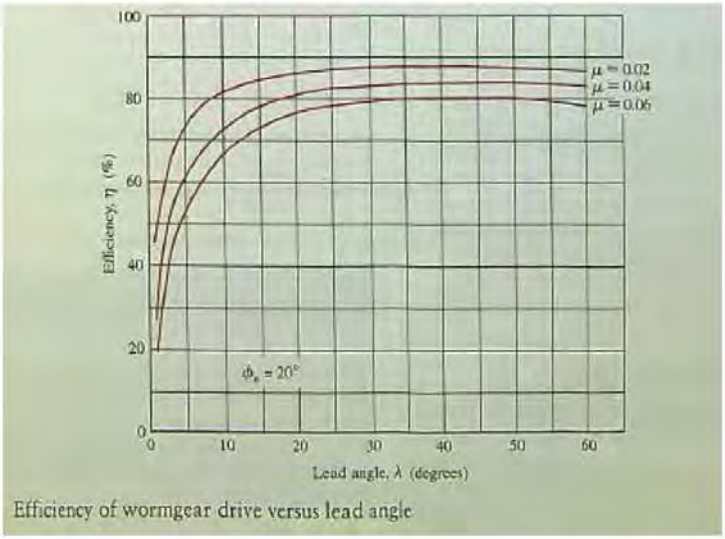
FIGURE 10–25 Coefficient of friction versus sliding velocity for steel worm and bronze wormgear

Eq. (15-38): $$f = 0.103\exp\left[-0.110(679.8)^{0.450}\right] + 0.012 = 0.0250$$

We can confirm the gear set will be "self-locking" by checking the condition on Equation 15-56. For steel worm on bronze gear the static friction factor is 0.15 (Figure 10-25). Equation 15-56 is for the gear to drive the worm so the locking condition is the opposite:

$f_{static} > \cos\phi_n \tan\lambda = \cos 20° \tan 4.764° = 0.078 < 0.15$ so we can expect self-locking

The mesh efficiency, e or η, can be found on the graph in Mott (Figure 10-26) which relates the output power to the input power. The derivation of e (or η) is the ratio of the transmitted force without friction to the force with friction.



Efficiency of wormgear drive versus lead angle

Eq. (15-54):

$$e = \frac{\cos\phi_n - f\tan\lambda}{\cos\phi_n + f\cot\lambda} = \frac{\cos 20° - 0.0250\tan 4.764°}{\cos\ 20° + 0.0250\cot 4.764°} = 0.7563 \quad Ans.$$

Eq. (15-58): $$W_G^t = \frac{33\,000 n_d H_o K_a}{V_G e} = \frac{33\,000(1)(1)(1.25)}{56.45(0.7563)} = 966 \text{ lbf} \quad Ans.$$

Eq. (15-57): $$W_W^t = W_G^t \frac{\cos\phi_n \sin\lambda + f\cos\lambda}{\cos\phi_n \cos\lambda - f\sin\lambda}$$

$$= 966\left(\frac{\cos 20°\sin 4.764° + 0.025\cos 4.764°}{\cos 20°\cos 4.764° - 0.025\sin 4.764°}\right)$$

$$= 106.4 \text{ lbf} \quad Ans.$$

An efficiency of 75% is quite low. This will also confirm that we can expect the gear set to be "self-locking." Increasing the worm lead angle as per the AGMA recommendations may improve the performance of the gear set.

Part 3: To be demonstrated by the TA.

c. Are the selected gears sufficient to handle the loading?

- We will use the AGMA standards for worm gear surface durability. This method from AGMA gives a method for rating the surface durability of hardened steel worms operating on bronze gears.
- The method provides for a maximum allowable tangential force
- The ratings are based on minimum wear and pitting on the gear
- Since the gear is made from bronze, it will wear faster than the worm

The equations in Shigley are shown as figures in Mott that highlight the relationships.

$$\left(W^t\right)_{\text{all}} = C_s D_m^{0.8} F_e C_m C_v \qquad \textbf{(15 - 28)}$$

where C_s = materials factor

D_m = mean gear diameter, in

F_e = effective face width of the gear (actual face width, but not to exceed $0.67 d_m$, the mean worm diameter), in

C_m = ratio correction factor

C_v = velocity factor

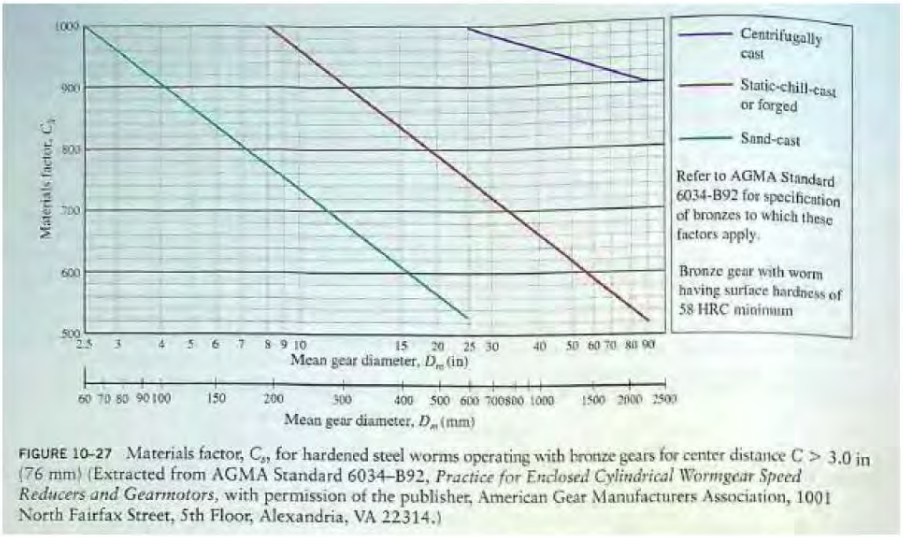


FIGURE 10–27 Materials factor, C_s, for hardened steel worms operating with bronze gears for center distance $C > 3.0$ in (76 mm) (Extracted from AGMA Standard 6034–B92, *Practice for Enclosed Cylindrical Wormgear Speed Reducers and Gearmotors*, with permission of the publisher, American Gear Manufacturers Association, 1001 North Fairfax Street, 5th Floor, Alexandria, VA 22314.)

Eq. (15-33): $C_s = 1190 - 477 \log 7.0 = 787$

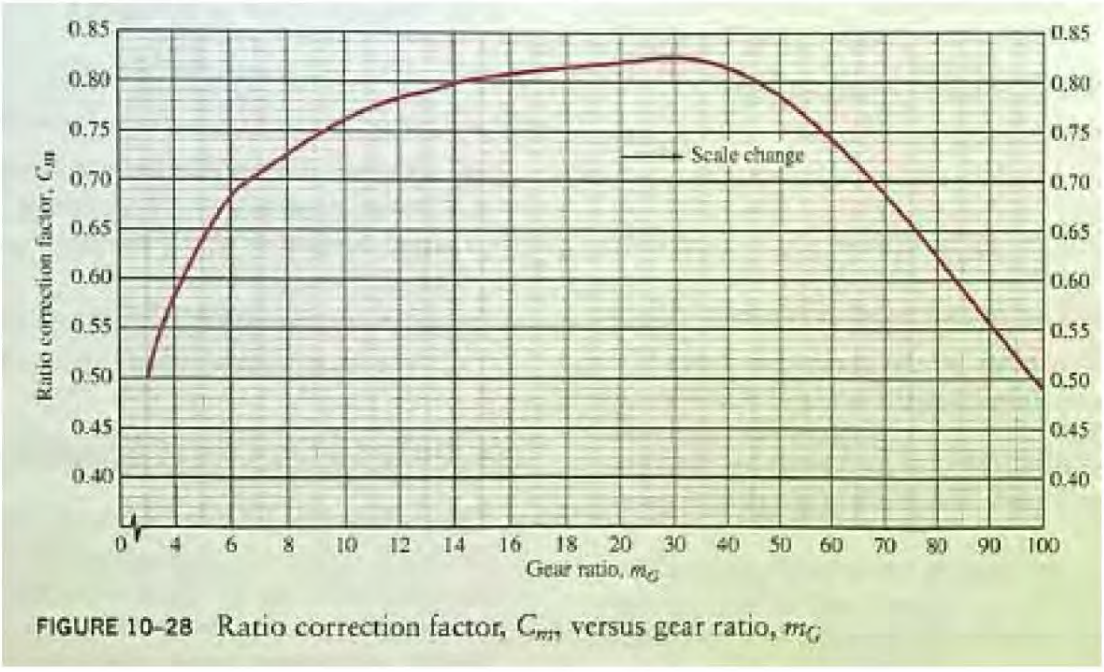


FIGURE 10–28 Ratio correction factor, C_m, versus gear ratio, m_G

Eq. (15-36): $$C_m = 0.0107\sqrt{-56^2 + 56(56) + 5145} = 0.767$$

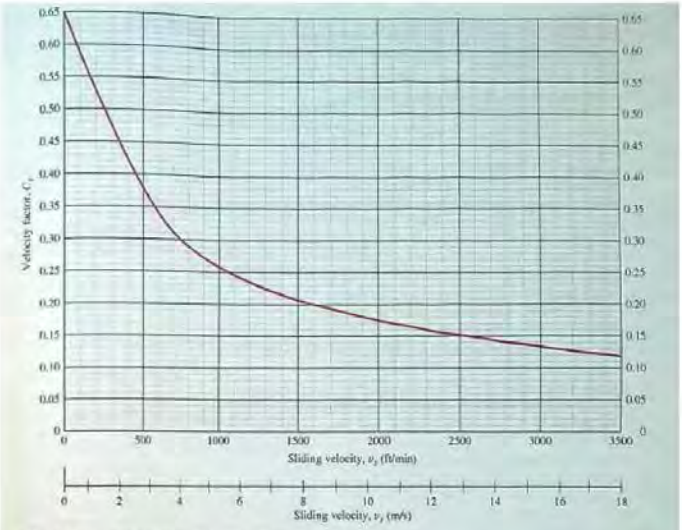


FIGURE 10-29 Velocity factor, C_v, versus sliding velocity

Eq. (15-37): $C_v = 0.659\exp[-0.0011(679.8)] = 0.312$

Eq. (15-28): $(W^t)_{all} = 787(7)^{0.8}(2)(0.767)(0.312) = 1787$ lbf

Since our design transverse force on the gear (966 lbs) is less than the allowable $W_G^t < (W^t)_{all}$, the mesh will survive at least 25 000 h.

The friction force that is acting against the worm action can be found from 15-61 and translating this to power using the sliding friction determined in part a. (V_s = 679.8 fpm).

Eq. (15-61): $$W_f = \frac{0.025(966)}{0.025\sin 4.764° - \cos 20°\cos 4.764°} = -29.5 \text{ lbf}$$

Eq. (15-63): $$H_f = \frac{29.5(679.8)}{33\,000} = 0.608 \text{ hp}$$

The power input to the worm is worm tangential speed applied to the transmitted force, W_W^t while the gear has the similar expression:

$$H_W = \frac{106.4(677.4)}{33\,000} = 2.18 \text{ hp}$$

$$H_G = \frac{966(56.45)}{33\,000} = 1.65 \text{ hp}$$

The gear power is higher than the expected output power of 1.5HP. This is expected as the service (application factor) of 1.25 is applied to determine the gear transverse force W_G^t = 966 lbs. The ratio of 2.18/1.65 = 0.7569 which matches the efficiency in part b. The input power of 2.18HP minus the friction power of 0.608HP gives a net output of 1.57HP which is closer to the expected output. Hence the mesh is sufficient *Ans.*

Using the stress method of AGMA to determine the stress in the gear teeth, we can

use Equation 15-53. A better explanation of the method is found in Mott section 10-11. For the lewis factor for the gear teeth, one can use y = 0.125 for φ = 20° (see bottom of page 817).

$$\sigma_a = \frac{W_G^t}{p_n F_e y} \qquad \textbf{(15-53)}$$

$$P_n = P_t / \cos\lambda = 8 / \cos 4.764^\circ = 8.028$$

$$p_n = \pi / 8.028 = 0.3913 \text{ in}$$

$$\sigma_G = \frac{966}{0.3913(0.5)(0.125)} = 39\ 500 \text{ psi}$$

The stress is very high. Using the same method from Mott but this time the transverse force is modified by a dynamic load factor, K_v.

Mott equation (10-37) $\sigma_a = W_d / yFp_n$ with

Dynamic load $W_d = W_G^t / K_v$ and $K_v = 1200 / (1200 + v_{tG})$

$K_v = 1200 / (1200 + 56.45) = 0.955$
Dynamic load $W_d = 966 / 0.955 = 1011$ lbs.

$\sigma_a = 1011/(0.125)(0.5)(0.3913) = 41{,}357$ psi

This value exceeds the rating for the fatigue strength of phosphorous bronze gears given as $\sigma_a = 24000$ psi. Hence we should reject this gear design.

One can look at the value of the stress at the actual rating of 1.0HP rather than the effective value of 1.65HP calculated above. In this case we obtain a value that just meets the allowable limit.

$$\sigma_G = \frac{1}{1.65} 39\ 500 = 23\ 940 \text{ psi} \quad \text{acceptable}$$

d. Estimate the lubricant sump temperature assuming normal convective heat transfer

This analysis requires us to use the expressions on page 817. Since we are given that the side area is 850in^2 we can assume that total lateral area is 1700in^2.

Heat loss rate from worm-gear case in ft·lbf/min,

$$H_{loss} = 33\,000(1-e)H_{in} \qquad \textbf{(15-49)}$$

Overall coefficient for combined convective and radiative heat transfer from the worm-gear case,

$$h_{CR} = \begin{cases} \dfrac{n_W}{6494} + 0.13 & \text{no fan on worm shaft} \\ \dfrac{n_W}{3939} + 0.13 & \text{fan on worm shaft} \end{cases} \qquad \textbf{(15-50)}$$

With case lateral area A, the oil sump temperature,

$$t_s = t_a + \frac{H_{loss}}{h_{CR} A} = \frac{33\,000(1-e)(H)_{in}}{h_{CR} A} + t_a \qquad \textbf{(15-51)}$$

AGMA recommended minimum lateral area in in^2

$$A_{min} = 43.20C^{1.7} \qquad \textbf{(15-52)}$$

Eq. (15-52): $A_{min} = 43.2(8.5)^{1.7} = 1642 \text{ in}^2 < 1700 \text{ in}^2$

Eq. (15-49): $H_{loss} = 33\,000(1 - 0.7563)(2.18) = 17\,530 \text{ ft} \cdot \text{lbf/min}$

Assuming a fan exists on the worm shaft. **Note: Better to assume no fan on the gearbox**

Eq. (15-50): $h_{CR} = \frac{1725}{3939} + 0.13 = 0.568 \text{ ft} \cdot \text{lbf/(min} \cdot \text{in}^2 \cdot {}^\circ\text{F)}$ ≈ 0.396

6494

Eq. (15-51): $t_s = 70 + \frac{17\,530}{0.568(1700)} = 88.2^\circ\text{F}$ *Ans.* ≈ 96.0°F

0.396

Since we have a temperature of 96°F (36°C), we are well below the temperature for lubricants of 250°F (120°C)

Part 4: To be done by the students.

e. Will a change in the gear width (face width) allow the worm gear box better meet the AGMA requirements? Is so, what do you suggest?

As shown in part c) the bending stress is very high. We can set the allowable stress at 24ksi and solve for Face width F.

Mott equation (10-37) $\sigma_a = W_d / yFp_n$ with

Dynamic load $W_d = W_G^t / K_v$ and $K_v = 1200 / (1200 + v_{tG})$

$K_v = 1200 / (1200 + 56.45) = 0.955$
Dynamic load $W_d = 966 / 0.955 = 1011$ lbs.

$\sigma_a = 1011/(0.125)(F)(0.3913) = 24000$ psi

F = 0.86inches.

We also need to restrict face width such that $F_e < 0.67d_m$. With worm diameter as 1.5 inches, we have $F_e < 1.0$ inches. The best option is a width of 7/8 inch = 0.875 inch

σ_a = 1011/(0.125)(0.875)(0.3913) = 23622psi

This value gives a safety factor of 1.02 for the design. exceeds the rating for the fatigue strength of phosphorous bronze gears given as σ_a = 24000 psi.

However, when we return to the AGMA wear equation (15-28) we find that that allowable tangential load still remains below the design load.

Eq. (15-28): $(W^t)_{all} = 787(7)^{0.8}(0.875)(0.767)(0.312) = 782$ lbf < 966 lbf

The overall conclusion is to increase the worm size to say 2 inches so that face width can be as much as 1.5" which should provide adequate allowable loads.